Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Increasing the Efficiency of Exercise Countermeasures for Bone Loss		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baldwin, Kenneth (University of California, Irvine) Hogan, Harry (Texas A&M University)		
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Task Description:	The over-arching goal of the proposed research is to test the hypothesis that astronaut exercise time designed to mitigate bone loss can be reduced if combined with an anti-resorptive pharmacological agent. The specific objectives are to define an optimal exercise regimen that will minimize bone loss during prolonged unloading and then to test reduced doses of that exercise regimen in combination with alendronate administration. Importantly, skeletal muscle function, mass, and phenotype will be simultaneously tested to assure that any reductions in exercise time do not sacrifice benefits to muscle. These experiments utilize the well-accepted model of hindlimb unloading (HU) in mature adult rats. Specific aim #1 will test two different modes of loading (electrically stimulated muscle contractions and voluntary resistance exercise) for their effects on mitigating bone and muscle decrements during 28 days of HU. The regimen judged most successful in mitigating bone loss will be further refined in specific aim #2, by testing varying frequency or session durations during HU. In addition, a dose-response study using alendronate during HU will be performed. That volume of training producing the maximal benefit to bone and acceptable benefit to muscle will then be tested in combination with the minimal dose of alendronate producing significant attenuation of bone loss. Specific aim #3 will test systematically reduced volumes of the optimized loading regimen in combination with alendronate to determine what reduction in exercise time can be achieved without sacrificing benefits to bone and muscle. A major focus during year 2 was to evaluate the efficacy of controlled muscle stimulation producing high-force eccentric contractions (simulating resistance exercise) as a countermeasure for the deleterious effects of hindlimb unloading (HU) on muscle and bone in rats. This training was performed 3 days/week during HU, utilizing 4 sets of 10 contractions at 120% of peak isometric torque. After 28 d, total vBMD at the proximal tibia (a site sensitive to disuse) was 11% higher in the HU-exercised (right) tibia compared to the HU-unexercised (left) leg. Mechanical testing of cancellous bone strength in the proximal tibia of the exercised leg in HU rats revealed that ultimate stress and elastic modulus, important material properties, were statistically identical to those in BL and CC weight bearing rats and significantly higher vs. the HU-unexercised leg. These results demonstrate the efficacy of the exercise protocol in preventing the substantial deterioration of bone mechanical integrity induced by HU. The most exciting and novel outcomes of this year's work are the strain data collected during acute studies performed during hindlimb unloading, characterizing the loading bone experiences during our resistance training paradigm (stimulated high-force muscle contractions). Strain gages were implanted on the anterio-medial surface of the proximal tibia to characterize the mechanical loading on bone during stimulated muscle contractions on Days 0 (baseline), 14, and 21 of HU. Interestingly, bone strain decreased with time (i.e., duration of HU) at the same muscle torques; further data analysis will be required to determine the contributions of any changes in bone vBMD or muscle torque to this reduction in strain magnitudes. Strains at the target exercise level (120% of peak isometric torque) ranged from 600 to 1200 microstrain, verifying that this training paradigm is effective in producing strain levels that have previously been proven to be osteogenic. Work continues on the other resistance training paradigm we wish to test during HU: voluntary exercise on a flywheel device that can be performed with the animal in the head-down tilt posture, maintaining the headward fluid shifts that we believe are an essential part of this unloading model. We are revising the tail harness used in these experiments in order to facilitate a more effective rat-machine interface to achieve the optimal training stimulus to hindlimb bone. As soon as this change in protocol is ironed out, we will perform HU experiments with flywheel training performed every other day, which allows us to complete Specific Aim #1. We are already moving forward to Specific Aim #2, in testing variations in the muscle stimulation protocol to achieve optimal mitigation of both bone and muscle atrophy. Because our first protocol was relatively ineffective in mitigating muscle loss and in stimulating muscle anabolic responses, we are modifying the parameters of muscle stimulation (including range of motion at the ankle, and intensity, duration of contractions) to test a new “dose” of simulated resistance training. We are working on the viability of chronic strain gage implants which would enable measurement of bone strain over a number of consecutive days and in the conscious animal performing voluntary flywheel exercise. Finally, as soon as we can acquire alendronate, we will be performing the dose-response experiments with alendronate during 28-d HU. The goal is to complete Specific Aim #2 by the end of this project year: testing our optimal dose of alendronate together with our optimized resistance training protocol for minimizing loss of bone and muscle functional integrity. These integrated experiments address five critical questions on the current Critical Path Roadmap relevant to bone health and skeletal muscle function. They will also provide fundamental and unique information about the biological interaction of these two tightly integrated tissues in response to the anabolic stimulus of exercise and the anti-resorptive effects of alendronate. The outcomes of these animal protocols can be easily translated to human studies using appropriate ground- and flight-based test beds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our detailed dose-response studies of the bone response to resistance training applied as a countermeasure during a period of simulated microgravity will help define the training stimulus required for maintaining bone health during prolonged periods of bed rest in clinical patients or others restricted to bed rest. Subsequent trials will test the efficacy of alendronate, an FDA-approved anti-resorptive agent commmonly prescribed for osteoporotic patients, versus resistance training versus combined alendronate and exercise in maintaining bone health during a period of unloading. These studies will answer whether there is an additive or synergistic effect of these pharmacological and physiological approaches. Given the numbers of Americans currently taking alendronate and other closely related bisphosphonate agents who also exercise, this could prove to be very useful data that could be used effectively to predict the human response to the combined effects of exercise and bisphosphonate treatment. Finally, our experiments testing various durations of exercise in combination with alendronate will quantify the minimum volume of exercise required to achieve the desired outcome of maintaining bone mass during prolonged periods of non-weightbearing when combined with this anti-resorptive treatment. This has important implications for rehabilitation specialists, physical therapists, and any clinicians treating bedridden patients. These research results are most relevant to humans flying in space, given that our rodent model mimicks both the mechanical unloading of the lower limbs and the fluid shifts observed with spaceflight. However, these results will also be pertinent to ground-based populations. There are limited studies to date on the efficacy of exercise combined with alendronate on bone phenotype outcomes, and none to our knowledge has pursued the more detailed dose-response studies proposed with this project. If indeed we can successfully demonstrate that simultaneous use of resistance training and alendronate is effective in minimizing bone loss with simulated microgravity, this provides additional support to the concept that exercise combined with anti-resorptive pharmacological agents may be helpful in preventing or attenuating

	the slower bone loss observed with aging. These results would also be of high interest to physicians and physical therapists in rehabilitation medicine, with reference to patients recovering from prolonged immobilization following complicated injuries or from prolonged bed rest.
Task Progress:	Significant progress has been made in identifying an optimal resistance training paradigm for mitigating bone and muscle loss in a rodent hindlimb unloading model. 1. Performance of a pilot study to test the effectiveness of our muscle stimulation protocol in normally weightbearing rats: Adult male SD rats were exposed to six training sessions using fine percutaneous needles (protocol from Ken Baldwin's lab) over 2 weeks (4 sets of 10 contractions at 120% peak isometric torque, 3 days/week). Frozen muscle samples (medial gastrocnemius) shipped to Co-I KM Baldwin's lab for assays of molecular markers indicating whether anabolic response achieved. 2. Completion of first full HU + stimulated muscle contraction experiments, n = 24 rats (12 HU, 1 leg stimulated, the other untreated; 12 aging cage controls). This required 3 months, running 12 animals at any one time due to the labor-intensive nature of the muscle stimulation protocols. Training was performed 3 d/week during 28 d HU. Outcomes: muscle functional properties, pQCT (pre-post) measures of vBMD and cross-sectional geometry on proximal and mid-shaft tibia, RPC testing on proximal tibia for mechanical integrity, muscle weights. Frozen medial gastrocnemius samples were sent to K. Baldwin's laboratory for gene expression assays . Bone samples are stored for histomorphometric studies of bone formation rates and cancellous bone microarchitecture. 3. Characterization of bone strain during stimulated muscle contractions: we performed a companion study with the HU + resistance training experiments described above to characterize bone strains incurred during these stimulated muscle contractions. Acute experiments were performed on rats on Days 0 , 7 and 21 d of HU (n=3 at each timepoint). Important information was gained on the magnitude and dynamic characteristics of the loading stimulus to bone with this training paradigm, and, equally important, data on how this loading stimulus changes as bone and muscle change over the course of HU. 4. Development of our other training paradigm (voluntary resistance exercise on a flywheel device) continues: we continue to refine procedures for operant conditioning of rats required before HU experiments can begin. In addition, we are modifying the tail harness used for achieving tail suspension, using a protocol successfully used by our consultant, Dr. Jim Fluckey, to enable a more effective rat-machine interface during flywheel training. 5. Investigator meetings were held at the NSBRI Retreat in February, 2006; frequent discussions about data and experimental design improvements occur via e-mail communications. Consultant Dr. Jim Fluckey, formerly of University of Arkansas Medical Sciences, assumed a faculty position at Texas A&M University, facilitating more rapid progress on the flywheel resistance training component of the project.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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Awards	Alcorn J. "Houston Society for Engineering in Medicine and Biology, Feb. 9-10, 2006, Houston, TX; to GRA Justin Alcorn." Feb-2006
Awards	Swift J. "Texas Chapter, American College of Sports Medicine, 1st Place MS Student Research Poster Award to GRA Joshua Swift, February 2006." Feb-2006
Awards	Nilsson M. "Texas Chapter, American College of Sports Medicine, 1st Place Ph.D. Student Research Poster Award to GRA Mats Nilsson, February 2006." Feb-2006